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Operationalization of Regenerative Design Indicators: An Integrated Framework of Design and Analysis

11

Arlind Dervishaj

Abstract

Different frameworks, sustainable design theories, and green building rating systems have been developed to reduce environmental impact on the planet. However, environmental challenges are growing due to climate change. The impact of the built environment on the planet requires an immediate transformation to achieve the sustainable development goals (SDG) and Paris Agreement. This study reviews some key concepts of regenerative design (RD), and how this can be integrated with state-of-the-art digital design methods. Even though building information modelling and design computation are of interest, their use and discussion have focused on sustainability rather than regeneration. Thus, there is a gap when it comes to their use for regenerative design. For this purpose, an integrated design and analysis framework is proposed where methods, processes, and tools can be used to measure indicators for regenerative design. The framework can be applied to case studies by researchers or designers and further expanded in future research.

Keywords

Design framework · Climate adaptation · Architecture · Metrics · Parametric design

11.1 Introduction

The built environment is responsible for 36% of global final energy consumption and approximately 37% of energy-related CO_2 emissions (Global Alliance for Building and Construction 2021). Urbanization trends are driving the need for new construction and that is predicted to double the building stock up to 2060, (International Energy Agency and the United Nations Environment Programme 2018). The European Union (EU) aims to become the first climate-neutral continent by 2050, and European Green Deal set the roadmap for a sustainable economy with net zero balance CO_2 emissions. In Europe, buildings are responsible for around 40% of final energy consumption and 36% of energy-related CO_2 emissions (European Commission).

The Intergovernmental Panel on Climate Change (IPCC) reports show that human activities have caused approximately a 1.0°C of global warming compared to pre-industrial levels (Intergovernmental Panel on Climate Change 2018; United Nations 2015a). Limiting global warming means reducing global emissions from human activities. The Carbon Budget, that is the

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additional carbon entering the atmosphere causing a 1.5°C global warming, was expected to be reached within ten to fourteen years. However, a recent report by the World Meteorological Organization already predicts a 50% probability of temporarily surpassing the IPCC's special report target of 1.5 any year from 2022 to 2026 (World Meteorological Organization 2022).

Given the evidence and the impact of the construction sector, the built environment can play a significant role in adaptation to climate change and achieving the SDGs. The sector needs to drastically reduce carbon emissions (contributing to SDG 11 Climate action), enabling circular approaches and reusing the existing building stock (SDG 12 responsible consumption and production), and adopting digital technologies (SDG 9 industry innovation and infrastructure) that have been argued as key to the sustainable transition to achieve the SDGs (Bai et al. 2020; United Nations 2015b).

11.1.1 Overview of Sustainable Design

Throughout human history, architecture has been a result of collaboration among more and more people, local climate and available materials, cultural contexts and needs, building knowledge and traditions, as well as codified styles. New societal developments change our understanding of what architecture's purpose is and how it is produced. For example, several new typologies of the nineteenth century were not designed by architects and faced strong opposition, and the addition of ornamental features was considered necessary, e.g. train stations. Currently, the Fourth Industrial Revolution, climate change issues, and the need for more sustainable approaches are expected to transform many industries, as well as the construction sector (Bai et al. 2020). Requirements for buildings change and develop at a fast pace in our times. In addition, several green building rating systems (GBRS) such as LEED, WELL, and DGNB have an international reach but represent voluntary certifications that can assist to design, build, and operate high-performance buildings (German Sustainable Building Council 2020; IWBI 2022; U.S. Green Building Council 2021). Even though the sustainability concept is comprised of three parts that are the social, financial, and environmental dimensions (Purvis et al. 2019), the environmental aspects (mainly on building performance) have been predominant in such standards, compared to the economic or social dimensions (Capolongo et al. 2015). Although green buildings have more ambitious goals compared to regulations, this is changing, as new and upcoming regulations require reporting of the embodied or whole life cycle carbon emissions (e.g. Climate declarations in Sweden), or setting an upper limit such as in France in the new regulation RE2020 (Boverket 2020; Ministry of the Ecological Transition (France) 2020). Moreover, sustainability and green building have not been enough to provide a safe and just space for humanity to thrive within planetary boundaries, or undo damage to ecosystems, aiming only for reduced impact (Dearing et al. 2014).

11.1.2 Background on Regenerative Design

Regenerative design and development (RDD) is a disciplinary study field since its commencement in 1994 by John Tillman Lyle (Mang and Reed 2013). It still consists of a few proponents and built examples. It has been developed in practice and theory in the pioneering work of "The Regenesis Group" for two decades. Additionally, it is seen of interest in more fields adopting the regenerative paradigm such as agriculture, city planning, fashion, and society and culture (Pedersen et al. 2022; Wahl 2016). Regenerative development represents an important shift for the needed transformation of the built environment and human activities by demanding living and whole systems thinking from initial planning to the design and actual functioning of human and natural systems. Furthermore, the definition of sustainable development, "*a development that meets the needs of the present without compromising the ability of*

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future generations to meet their own needs", and the subsequent approaches to sustainable design, are not only human-centric but have proven to be insufficient to climate change mitigation and adaptation, and biodiversity loss (International Panel on Climate Change 2022; World Commission on Environment and Development 1987). Sustainability thinking relies on doing nothing or a few actions to prevent damage to ecological systems from human exploitation.

Sustainable or green building practices have focused on reducing negative impacts on the environment, providing indoor comfort for people, and increasing efficiency in the use of resources and processes (Reed 2007). Regenerative design aims to go beyond the current slow degeneration of planetary resources and ecosystems, towards reversing the damage and restoring restore nature to a healthy state (Cole 2012). Regenerative practices work towards designing buildings and cities that strive to give more than they take in the local place, supporting the co-evolution of both human and natural systems. The regenerative approach requires holistic thinking in the planning/design process and can contribute to the SDGs such as 11 sustainable cities and communities, 14 life below water, and 15 life on land.

Different concepts and principles of sustainable design have been investigated to support regenerative design (Naboni and Havinga 2019). Regeneration has been argued to combine a confluence of principles from biophilia, biomimicry, climate-neutral buildings, and more, to create environments that restore ecosystems and support human health and well-being (i.e. SDG 3), and bring benefits for both natural and human systems (Mauree et al. 2019; Naboni and Havinga 2019; Zari 2012). Some of the principles of regenerative design discussed in the professional and academic literature are as follows:

- Carbon-positive buildings, e.g. use of bio-based materials, reuse of existing buildings, reversible design, and reuse at end-of-life.
- Plus energy buildings, e.g. low energy needs and use of renewable energy
- Climate adaptation and reverse ecosystem degradation
- Circular design and construction objectives for all life cycle stages and resources.

Adapted from Reed, Fig. 11.1 provides an updated overview of where different initiatives stand including conventional practice, green building, sustainability, and restorative towards regenerative. The best that conventional or sustainable design does is to improve systems efficiency and to reduce the environmental impact of degenerating natural systems, aiming for neutrality. The restorative design seeks to bring back human and natural systems to a healthy state while the regenerative principles strive to make

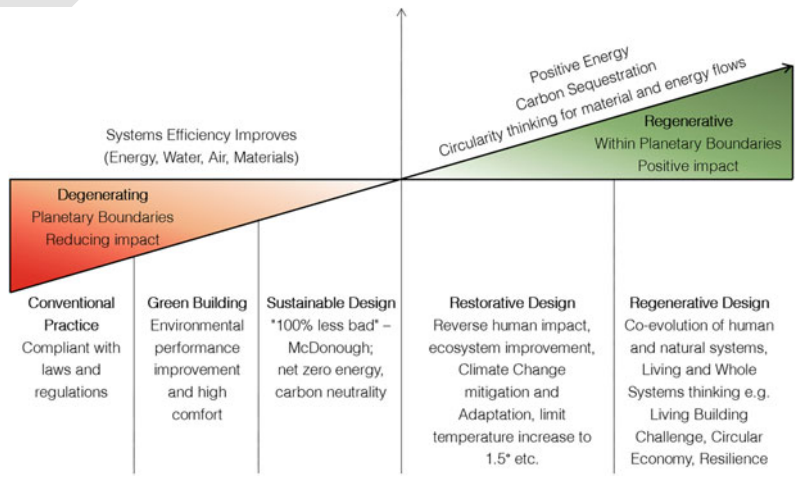


Fig. 11.1 Regenerative development and design trajectory, adapted from Reed (2007)

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developments where social and ecological systems can reach a healthy state.

Green certifications, for example, the Living Building Challenge, are shifting from reducing impact towards buildings that have a positive impact in terms of energy and carbon in the building site (International Living Future Institute (ILFI) 2020). Moreover, designing with climate, designing with nature, and designing with people have been discussed as three dimensions of regenerative buildings (Naboni and Havinga 2019). "Design with Climate" is also the title of a book introducing bioclimatic design which is a systematic approach to design based on local climate for human comfort (Olgyay 1963). "Design with Nature" is another book that introduces several concepts of ecological planning, holistic thinking, and integrating natural processes into planning and design. In addition to the above, we need to "design with people" to achieve restorative and regenerative whole systems. Connection to nature, both outdoor and indoor environments, for example through windows and elements that determine the quality of the view such as content, access, and clarity, play an important role in human well-being and productivity in the workplace and have long-term health benefits (Ko et al. 2022).

11.1.3 Research Objectives

RD theory has been developed over a long timeframe by a few authors. Some recent researchers study RD through case studies and investigating digital workflows and methods such as daylight, energy and life cycle assessment (LCA) to achieve positive outcomes (Gardner et al. 2020; Naboni et al. 2019). Although building information modelling (BIM) and computational design have been studied thoroughly such as regarding sustainability as a further dimension in BIM, green buildings, or building performance simulations (Charef et al. 2018; Koutamanis 2020), there is a gap when it comes to their use for RD. Therefore, this paper aims to investigate digital methods that are relevant to RD and discuss the operationalization of indicators, meaning turning RD goals into measurable metrics, by capturing design objectives in digital processes by collecting and proposing metrics that can contribute to positive outcomes.

The following research questions are addressed:

- What are some digital methods that can assist in regenerative design?
- How to operationalize performance indicators using an integrated framework of design and analysis for regenerative development and design?

11.2 Methods

In this section, an overview of analogue and digital design methods is given. Bioclimatic design is described with regard to RD. Digital design methods such as computer-aided design (CAD), BIM, and computational design are utilized and considered relevant for RD. Digital tools were collected from the existing and recent scientific literature discussing RD in combination with digital methods. Additionally, some methods and tools used in the case studies of the regenerative design handbook are included (Naboni and Havinga 2019). The methods and tools are useful to apply regenerative design at an urban (e.g. masterplan) and building design scale.

11.2.1 Analogue and Digital Methods for Regenerative Design

Typically, designers begin to study a project by understanding the context and using drawings and sketches to explore ideas. Additionally, tacit knowledge acquired through many years of practice plays an important role in the thinking and subsequent design choices. Scale models and analogue analysis, such as using a heliodon (a device to simulate sun and shadow with artificial light) or placing the model outdoors in the sun for analysis, are still part of architecture curricula and professional practice. Standards on indoor

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comfort (e.g. ISO 7730, ASHRAE 55:2017, and EN 16798–1:2019) and regulations on sunlight and daylight (e.g. EN 17037) can contribute to a regenerative indoor environment. Moreover, bioclimatic design charts are used to select passive and active strategies for achieving thermal comfort. The sun charts and the wind rose guide design strategies in the local climate and context. The positive impacts of bioclimatic methods such as the use of local materials and vernacular solutions may contribute to low-carbon buildings and reduced energy demands from using natural energy flow such as daylight, natural ventilation, and solar energy (Watson 2013). Therefore, bioclimatic approaches are highly relevant for RD to reduce environmental impacts and support human health. Furthermore, bioclimatic methods have become digitized and interactive, through digital tools such as "Climate Consultant", or part of design modelling tools such as Ladybug Tools. Daylight factors are being gradually substituted with climate-based metrics that can be visualized onto the model's geometry (IESNA-Daylight Metrics Committee 2013; European Committee for Standardization 2018). Some recent digital tools are used for modelling urban energy, microclimate, daylight, and planning indicators (e.g. floor area ratio and gross floor area). Tools like ENVI-met, City Energy Analyst, Urban Modelling Interface, and CitySim Pro are used to analyse at the urban scale. Additionally, other tools like Spacemaker, Hypar, and TestFit leverage APIs for GIS data integration and allow for fast modelling of design alternatives and near real-time computation of planning metrics. Simulations for energy, daylight and computational fluid dynamics (CFD) are usually performed with specialized software. There is a need for specialists to model and perform simulations in such cases. Validated simulation engines such as Radiance and EnergyPlus are used by many commercial and open-source simulation tools. In addition, some simulations can be performed inside BIM tools, but flexibility and control over parameters are lost, and it may be unsuitable beyond early design estimations, e.g. Automated Energy Model in Revit. Thus, flexible workflows that allow for user freedom are an ongoing research and development direction. Even though computational methods offer more flexibility and freedom, it requires specialized skills and time to develop workflows. As a result, large architecture/engineering firms have dedicated specialists working with computational methods. However, open-source developments are beneficial for both small and large firms and designers working towards regenerative goals.

In Fig. 11.2, a sample of tools and categorization based on design scale and design/analysis type is proposed. Programming (to build/modify tools) and validated simulation tools stand at the foundation on which regenerative design workflows can be developed. CAD and BIM tools are shown in one category as they are integrated with computational tools (e.g. Dynamo and Grasshopper), as well as having integrations among them e.g. several Grasshopper links to other tools, or Autodesk's ecosystem of tools. Analysis workflows are placed in one category as they have distinct modelling requirements from design modelling tools. In addition, urban-level simulation tools are located separately from parametric tools since there are no workflows that connect models and results to the next design stage or tools. However, parameters and indicators of one workflow and early stage can be useful for subsequent phases.

11.3 Results and Discussion

11.3.1 An Integrated Design and Analysis Framework

Computer-aided design (CAD) tools started to be available almost 40 years ago. However, CAD tools are being gradually substituted with BIM, computational and web-based tools. BIM stands for a complete description of buildings. Information models are used for design or analysis purposes. BIM provides functionalities that support several design stages. In addition, computational workflows go beyond BIM applications and can be used for both urban and building design and across design stages and scales.

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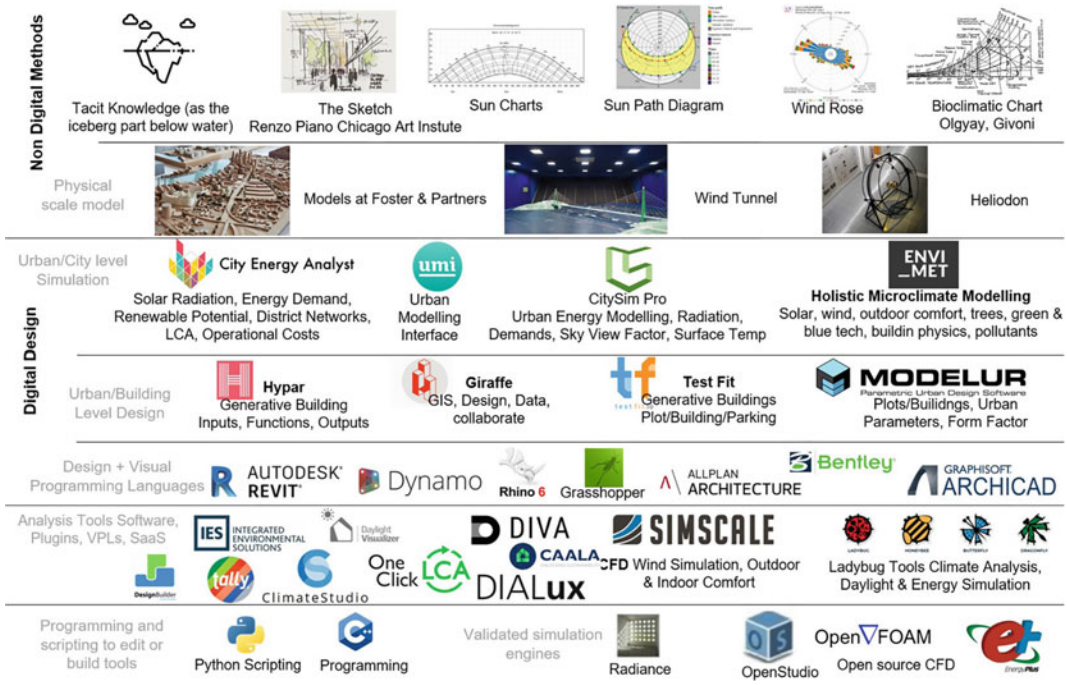


Fig. 11.2 Non-digital and some digital design and analysis methods relevant to RD

Stakeholders can benefit from indicators to make informed design choices. The design process is improved when performance feedback is given and visualized concurrently for a more comprehensive and holistic approach. For example, Naboni proposes a series of key performance indicators for regenerative urban design that are tested in a case study through a workflow by connecting algorithms and exchanging data within Grasshopper (Naboni et al. 2019). Visual programming languages (VPL) such as Grasshopper allow for parametric design of geometry and attaching attributes needed for operationalizing ways to measure design objectives. Such an integrated approach, when used concurrently in design and analysis iteratively, creates opportunities to investigate solutions that strive to achieve higher goals instead of typical performance trade-offs, e.g. on thermal comfort, microclimate, daylight, glare, energy, and environmental benefits. Therefore, digital methods can effectively support RD in the design process and provide feedback with measurable indicators.

11.3.2 Operationalizing Indicators for Regenerative Design

The previously discussed sections support an emerging framework that can serve to design and measure the performance of regenerative buildings. In Table 11.1, indicators are collected from academic literature, guidelines for cities and designers, and green building certifications discussing regenerative approaches. The indicators are categorized into the three distinct dimensions of climate, nature, and people for regeneration argued in Sect. 1.2. Table 11.1 represents an initial sample with some measurable indicators. Since the requirements of a design are specific to the local context, climate, legislation, and client brief, "operationalizing" (selecting and quantifying indicators) is a flexible methodology to set objectives and work collaboratively and develop workflows that capture design objectives and provide feedback. The climate indicators can be used to measure building performance. Even though climate and people indicators are related, the latter collects only indicators that measure the

Table 11.1 Indicators for a regenerative design framework based on climate, people, and nature dimensions

Climate (Performance)	People (Comfort and health)	Nature (Circularity of resources)
Daylight metrics, e.g. Illuminance (lux), daylight factor (%)	Glare, e.g. unified glare rating (UGR; %), daylight glare probability (DGP; %)	Nature-based solutions (urban forestry, sponge cities, and green infrastructure)
Climate-based daylight metrics, e.g. daylight autonomy (DA; %)	Climate-based annual glare, e.g. imageless DGP (Jones 2019)	Wastewater treatment
Minimum sunlight hours, e.g. Athens charter, EN 17037	Thermal comfort model, e.g. predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD)	Passive design strategies, e.g. trombe wall, internal heat gains, natural ventilation, and ground temperature vectors
Renewable energy production (kWh)	Adaptive model of thermal comfort	Biodiversity net gain
Annual energy loads (kWh)	Outdoor comfort based on the universal thermal climate index (UTCI)	Waste into a resource, e.g. heating or electric energy production
Energy use intensity (EUI) kWh/m^2/year	Pedestrian wind comfort (e.g. Lawson, davenport, NEN 8100)	Circular design extends the life of building materials through new life cycles
Embodied carbon ($kgCO_2eq/m^2$)	Air quality indicators	Designing out waste, e.g. 3Rs (reduce, reuse, and recycle)
Operational carbon ($kgCO_2eq/m^2$/year)	Water quality indicators	Rainwater collection and its reuse in buildings or landscape irrigation
Life cycle global warming potential or whole life carbon (GWP) $kgCO_2eq/m^2$/year	View quality, e.g. view out in EN 17037, window view quality assessment framework (Ko et al. 2021)	Low carbon materials and storage, e.g. bio-based materials such as wood and bamboo
Form factor, heat loss form factor (from the passive house institute)	Overheating hours	Biophilia, e.g. the introduction of natural elements into buildings
Life cycle assessment	Melanopic light intensity	Air and water quality for ecosystems and biodiversity

comfort and health aspects intending to support human well-being. For example, climate-based daylight metrics are part of the climate category, but metrics regarding glare (e.g. daylight glare probability or annual sunlight exposure), views (e.g. distance, clarity, content, and access), or healthy light exposure (e.g. melanopic lux) are included in the people category. Indicators under nature do not directly affect the indoor environment but are regarded as key for restorative and regenerative processes to take place through planning and building design, aiming towards a closed loop of resources. Additionally, applying some of the framework's methods and indicators requires the use of advanced tools and complex methods at present. Thus, progressive development of digital workflows, specialist/interdisciplinary skills, and collaboration of actors involved will be needed in specific cases.

The indicators in the nature section are not always quantifiable by metrics, or at the design stage by digital methods and may require years of measurement to evaluate. Cole discusses that regenerative development requires long timeframes and qualitative approaches to observe the positive benefits on natural and human systems (Cole 2012). Biodiversity net gain is a recent methodology developed in the United Kingdom that tries to quantify the outcomes by measuring the habitat's state pre and after-development

(Local Government Association). Another example of the people's category is the melanopic light intensity which measures the non-visual effects of light on humans responsible for keeping our circadian rhythms in sync with day and night cycles (Berson 2003). Circadian lighting design is a feature of the light concept in the WELL standard (IWBI 2022). When new research findings are presented, they raise our understanding of human well-being or the impacts on nature and biodiversity. Afterwards, methodologies and standards introduce scientific evidence that can contribute to more holistic design practice, thus a regenerative design process.

11.4 Conclusions

The need for more sustainable development within planetary boundaries is of paramount importance due to climate change and its implications. In this context, the built environment can play an important role in climate adaptation, reducing emissions as well as adopting regenerative approaches that are beneficial/positive for people and nature to thrive. Sustainability, circular economy, and green buildings still receive the most attention in research and practice. Digital methods have been previously developed and discussed for this purpose. However, there is some gap in how digital methods can contribute specifically to regenerative design and development.

This paper aimed to describe and discuss design and analysis methods, tools, and standards that are relevant to regenerative design. BIM and computational methods can support integrated digital processes that are argued as relevant to achieve a holistic design process and provide feedback loops for projects and stakeholders. In order to measure the performance of regenerative approaches, the operationalization of indicators needs to be captured in design processes supported by digital methods. For this purpose, a categorization of existing and emerging indicators is done based on climate, people, and nature dimensions identified for regenerative design by previous research. The indicators can contribute towards achieving SDG 3 good health and well-being, 7 affordable and clean energy, 11 sustainable cities and communities, 12 responsible consumption and production, 13 climate action, and 15 life on land and may relate indirectly towards the rest of the SDGs. Further research can investigate and expand the proposed methods and novel digital workflows applied to case studies for regenerative design. The proposed framework can be used by practitioners when working towards regenerative development and design goals with stakeholders.

We can no longer be concerned only with reducing our damage to the planet and reduced human health risks but should go beyond the Brundtland definition of sustainability to do better. Regenerative design shifts from reducing environmental impacts, towards creating positive outcomes for people and the planet. The regeneration and sustainment of all living systems are at the core of regenerative development and design.

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